

Commonly Used Taylor Series

SERIES	WHEN IS VALID/TRUE
$\frac{1}{1-x} = 1 + x + x^2 + x^3 + x^4 + \dots$ $= \sum_{n=0}^{\infty} x^n$	NOTE THIS IS THE GEOMETRIC SERIES. JUST THINK OF x AS r $x \in (-1, 1)$
$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$ $= \sum_{n=0}^{\infty} \frac{x^n}{n!}$	SO: $e = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \dots$ $e^{(17x)} = \sum_{n=0}^{\infty} \frac{(17x)^n}{n!} = \sum_{n=0}^{\infty} \frac{17^n x^n}{n!}$ $x \in \mathbb{R}$
$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \dots$ $= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$	NOTE $y = \cos x$ IS AN <u>EVEN</u> FUNCTION (I.E., $\cos(-x) = +\cos(x)$) AND THE TAYLOR SERIS OF $y = \cos x$ HAS ONLY <u>EVEN</u> POWERS. $x \in \mathbb{R}$
$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \dots$ $= \sum_{n=1}^{\infty} (-1)^{(n-1)} \frac{x^{2n-1}}{(2n-1)!} \stackrel{\text{or}}{=} \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$	NOTE $y = \sin x$ IS AN <u>ODD</u> FUNCTION (I.E., $\sin(-x) = -\sin(x)$) AND THE TAYLOR SERIS OF $y = \sin x$ HAS ONLY <u>ODD</u> POWERS. $x \in \mathbb{R}$
$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \dots$ $= \sum_{n=1}^{\infty} (-1)^{(n-1)} \frac{x^n}{n} \stackrel{\text{or}}{=} \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}$	QUESTION: IS $y = \ln(1+x)$ EVEN, ODD, OR NEITHER? $x \in (-1, 1]$
$\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \frac{x^9}{9} - \dots$ $= \sum_{n=1}^{\infty} (-1)^{(n-1)} \frac{x^{2n-1}}{2n-1} \stackrel{\text{or}}{=} \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$	QUESTION: IS $y = \arctan(x)$ EVEN, ODD, OR NEITHER? $x \in [-1, 1]$

Fix an interval I in the real line (e.g., I might be $(-17, 19)$) and let x_0 be a point in I , i.e.,

$$x_0 \in I .$$

Next consider a function, whose domain is I ,

$$f: I \rightarrow \mathbb{R}$$

and whose derivatives $f^{(n)}: I \rightarrow \mathbb{R}$ exist on the interval I for $n = 1, 2, 3, \dots, N$.

Definition 1. The N^{th} -order Taylor polynomial for $y = f(x)$ at x_0 is:

$$p_N(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \dots + \frac{f^{(N)}(x_0)}{N!}(x - x_0)^N , \quad (\text{open form})$$

which can also be written as (recall that $0! = 1$)

$$p_N(x) = \frac{f^{(0)}(x_0)}{0!} + \frac{f^{(1)}(x_0)}{1!}(x - x_0) + \frac{f^{(2)}(x_0)}{2!}(x - x_0)^2 + \dots + \frac{f^{(N)}(x_0)}{N!}(x - x_0)^N \quad \leftrightarrow \text{a finite sum, i.e. the sum stops .}$$

Formula (open form) is in open form. It can also be written in closed form, by using sigma notation, as

$$p_N(x) = \sum_{n=0}^N \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n . \quad (\text{closed form})$$

So $y = p_N(x)$ is a polynomial of degree at most N and it has the form

$$p_N(x) = \sum_{n=0}^N c_n (x - x_0)^n \quad \text{where the constants} \quad c_n = \frac{f^{(n)}(x_0)}{n!}$$

are specially chosen so that derivatives match up at x_0 , i.e. the constants c_n 's are chosen so that:

$$\begin{aligned} p_N(x_0) &= f(x_0) \\ p_N^{(1)}(x_0) &= f^{(1)}(x_0) \\ p_N^{(2)}(x_0) &= f^{(2)}(x_0) \\ &\vdots \\ p_N^{(N)}(x_0) &= f^{(N)}(x_0) . \end{aligned}$$

The constant c_n is the n^{th} Taylor coefficient of $y = f(x)$ about x_0 . The N^{th} -order Maclaurin polynomial for $y = f(x)$ is just the N^{th} -order Taylor polynomial for $y = f(x)$ at $x_0 = 0$ and so it is

$$p_N(x) = \sum_{n=0}^N \frac{f^{(n)}(0)}{n!} x^n .$$

Definition 2.¹ The Taylor series for $y = f(x)$ at x_0 is the power series:

$$P_\infty(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \dots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n + \dots \quad (\text{open form})$$

which can also be written as

$$P_\infty(x) = \frac{f^{(0)}(x_0)}{0!} + \frac{f^{(1)}(x_0)}{1!}(x - x_0) + \frac{f^{(2)}(x_0)}{2!}(x - x_0)^2 + \dots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n + \dots \quad \leftrightarrow \text{the sum keeps on going and going.}$$

The Taylor series can also be written in closed form, by using sigma notation, as

$$P_\infty(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n . \quad (\text{closed form})$$

The Maclaurin series for $y = f(x)$ is just the Taylor series for $y = f(x)$ at $x_0 = 0$.

¹Here we are assuming that the derivatives $y = f^{(n)}(x)$ exist for each x in the interval I and for each $n \in \mathbb{N} \equiv \{1, 2, 3, 4, 5, \dots\}$.

Big Questions 3. For what values of x does the power (a.k.a. Taylor) series

$$P_{\infty}(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n \quad (1)$$

converge (usually the Root or Ratio test helps us out with this question). If the power/Taylor series in formula (1) does indeed converge at a point x , does the series converge to what we would want it to converge to, i.e., does

$$f(x) = P_{\infty}(x) ? \quad (2)$$

Question (2) is going to take some thought.

Definition 4. The N^{th} -order *Remainder term* for $y = f(x)$ at x_0 is:

$$R_N(x) \stackrel{\text{def}}{=} f(x) - P_N(x)$$

where $y = P_N(x)$ is the N^{th} -order Taylor polynomial for $y = f(x)$ at x_0 .

So

$$f(x) = P_N(x) + R_N(x) \quad (3)$$

that is

$$f(x) \approx P_N(x) \quad \text{within an error of } R_N(x) .$$

We often think of all this as:

$$f(x) \approx \sum_{n=0}^N \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n \quad \leftrightarrow \text{a finite sum, the sum stops at } N .$$

We would LIKE TO HAVE THAT

$$f(x) \stackrel{??}{=} \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n \quad \leftrightarrow \text{the sum keeps on going and going .}$$

In other notation:

$$f(x) \approx P_N(x) \quad \text{and the question is} \quad f(x) \stackrel{??}{=} P_{\infty}(x)$$

where $y = P_{\infty}(x)$ is the Taylor series of $y = f(x)$ at x_0 .

Well, let's think about what needs to be for $f(x) \stackrel{??}{=} P_{\infty}(x)$, i.e., for f to equal to its Taylor series.

Notice 5. Taking the $\lim_{N \rightarrow \infty}$ of both sides in equation (3), we see that

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n \quad \leftrightarrow \text{the sum keeps on going and going .}$$

if and only if

$$\lim_{N \rightarrow \infty} R_N(x) = 0 .$$

Recall 6. $\lim_{N \rightarrow \infty} R_N(x) = 0$ if and only if $\lim_{N \rightarrow \infty} |R_N(x)| = 0$.

So 7. If

$$\lim_{N \rightarrow \infty} |R_N(x)| = 0 \quad (4)$$

then

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n .$$

So we basically want to show that (4) holds true.

How to do this? Well, this is where Mr. Taylor comes to the rescue!

²

²According to Mr. Taylor, his Remainder Theorem (see next page) was motivated by coffeehouse conversations about works of Newton on planetary motion and works of Halley (of *Halley's comet*) on roots of polynomials.

Taylor's Remainder Theorem

Version 1: for a fixed point $x \in I$ and a fixed $N \in \mathbb{N}$.³

There exists c between x and x_0 so that

$$R_N(x) \stackrel{\text{def}}{=} f(x) - P_N(x) \stackrel{\text{theorem}}{=} \frac{f^{(N+1)}(c)}{(N+1)!} (x - x_0)^{(N+1)} . \quad (5)$$

So either $x \leq c \leq x_0$ or $x_0 \leq c \leq x$. So we do not know exactly what c is but atleast we know that c is between x and x_0 and so $c \in I$.

Remark: This is a Big Theorem by Taylor. See the book for the proof. The proof uses the Mean Value Theorem. Note that formula (5) implies that

$$|R_N(x)| = \frac{|f^{(N+1)}(c)|}{(N+1)!} |x - x_0|^{(N+1)} . \quad (6)$$

Version 2: for the whole interval I and a fixed $N \in \mathbb{N}$.³

Assume we can find M so that

$$\text{the maximum of } |f^{(N+1)}(x)| \text{ on the interval } I \leq M ,$$

i.e.,

$$\max_{c \in I} |f^{(N+1)}(c)| \leq M .$$

Then

$$|R_N(x)| \leq \frac{M}{(N+1)!} |x - x_0|^{N+1} \quad (7)$$

for each $x \in I$.

Remark: This follows from formula (6).

Version 3: for the whole interval I and all $N \in \mathbb{N}$.⁴

Now assume that we can find a sequence $\{M_N\}_{N=1}^{\infty}$ so that

$$\max_{c \in I} |f^{(N+1)}(c)| \leq M_N$$

for each $N \in \mathbb{N}$ and also so that

$$\lim_{N \rightarrow \infty} \frac{M_N}{(N+1)!} |x - x_0|^{N+1} = 0$$

for each $x \in I$. Then, by formula (7) and the Squeeze Theorem,

$$\lim_{N \rightarrow \infty} |R_N(x)| = 0$$

for each $x \in I$. Thus, by So 7,

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n$$

for each $x \in I$.

³Here we assume that the $(N+1)$ -derivative of $y = f(x)$, i.e. $y = f^{(N+1)}(x)$, exists for each $x \in I$.

⁴Here we assume that $y = f^{(N)}(x)$, exists for each $x \in I$ and each $N \in \mathbb{N}$.